Linear Regression for Medical Insurance Charges Prediction

Executive Summary

This report presents the development of a linear regression model to predict individual medical insurance charges using demographic and lifestyle factors. Using the Kaggle Insurance dataset (1,338 records, 7 variables), the model achieved an R^2 of 0.78, explaining 78% of the variance in charges. Smoking status, BMI, and age were identified as the strongest predictors, with smokers incurring approximately $23,650 more in annual charges than non-smokers. The findings provide actionable insights for risk-based pricing and wellness programme targeting.

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1. Dataset Suitability (10 Marks)

1.1 Dataset Description

The **Insurance dataset** from Kaggle contains 1,338 records of individual medical insurance beneficiaries. It includes both demographic and health-related variables:

Age	Numerical	Age of primary beneficiary (18–64)
sex	Categorical	male / female
bmi	Numerical	Body Mass Index (15.96 – 53.13)
children	Numerical	Number of dependents covered (0–5)
smoker	Categorical	yes / no
region	Categorical	northeast, southeast, southwest, northwest
charges	Numerical (target)	Individual medical insurance bill

1.2 Suitability for Linear Regression

Linear regression is appropriate for this task because:

- **Target variable is continuous** –charges is a numerical amount, suitable for regression modelling.
- **Relationships are expected to be approximately linear** – prior research shows that age, BMI, and smoking have roughly additive effects on healthcare costs.
- **Interpretability is required** – coefficients directly indicate the dollar impact of each predictor, which is valuable for business decisions (e.g., premium setting).

1.3 Data Quality Assessment

- **Missing values**: None – all 1,338 rows have complete data across all 7 columns.
- **Data types**: Correctly assigned (numerical forage, bmi, children, charges; object for categorical variables).
- **Outliers**: charges are positively skewed (max = $63,770, mean = $13,270), with high-cost smokers as extreme values. This will affect MSE but does not invalidate linear regression.
- **Multicollinearity risk**: Initial correlations show moderate relationships (e.g., age vs charges~0.30), no extreme multicollinearity expected.

1.4 Potential Pitfalls Acknowledged

- Skewness in the target may violate the normality of residuals.
- Interaction effects (e.g., smoker x bmi) are not captured by a simple additive model.
- Outliers can disproportionately influence MSE.

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2. Planning & Exploratory Data Analysis (20 Marks)

2.1 Analysis Plan

Before coding, the following step-by-step plan was established:

1. **Data inspection** – Load data, check structure, identify missing values, and compute summary statistics.
2. **Exploratory Data Analysis (EDA)** – Visualise distributions, relationships, and correlations.
3. **Data preprocessing** – One-hot encode categorical variables (sex, smoker, region)
4. **Train-test split** – 80% training, 20% testing (random state = 42 for reproducibility).
5. **Model training** – Fit a linear regression model on the training set.
6. **Evaluation** – Compute MAE, MSE, R^2, and compare against a baseline (predicting the mean).
7. **Interpretation** – Analyse coefficients and discuss business implications.
8. **Diagnostics** – Residual plots (post-evaluation) to check assumptions.

2.2 Exploratory Data Analysis (EDA)

2.2.1 Dataset Overview

python
```
df.info()
df.describe()
```

- **Range**: Age 18–64, BMI 15.96–53.13, children 0–5, charges $1,122–$63,770.

- No missing values – clean dataset.

2.2.2 Distribution of Charges

Finding: The histogram of charges is positively skewed – most individuals have moderate costs (<$15,000), but a long right tail extends beyond $50,000.
Interpretation: The skewness suggests that a small group (predominantly smokers) incurs very high medical bills. This can affect the normality assumption of residuals; a log transformation may be considered in future iterations.

2.2.3 Smoking Status vs. Charges

Finding: The boxplot shows a stark difference – smokers have median charges around $30,000, while non-smokers have a median of approximately $8,000 (a ~275% increase).
Interpretation: Smoking is the strongest single predictor of insurance costs. Insurers should place heavy weight on smoking status in premium calculations.

2.2.4 Age vs. Charges

Finding: Scatter plot reveals a weak positive linear trend. Correlation coefficient ≈ 0.30.
Interpretation: Age alone explains only about 9% of the variance in charges. However, age interacts with other factors (e.g., older smokers have even higher costs).

2.2.5 BMI vs. Charges

Finding: Scatter plot shows a moderate positive relationship, with a correlation ≈ 0.20. The relationship is more pronounced among smokers.
Interpretation: Higher BMI is associated with higher medical costs, but the effect is relatively modest. This suggests that lifestyle interventions targeting BMI reduction could yield some savings, but less than smoking cessation.

2.2.6 Correlation Matrix

Smoker (encoded)	0.79
age	0.30
bmi	0.20
children	0.07

Interpretation: Smoking dominates the correlation structure. The weak correlation for children indicates that the number of dependents has little linear relationship with individual charges.

2.3 Key Insights from EDA

- **Smoking is the dominant predictor** – it drives most of the variation.
- **Charges are right-skewed** – outliers exist; MSE will be sensitive.
- **Age and BMI have positive but modest linear relationships**.

- **No missing values** – no imputation needed.
- **Relationships appear largely linear** – linear regression is a reasonable starting point.

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3. Conduct Analysis – Model Training (30 Marks)

3.1 Feature Selection

All available features were retained because:

- EDA indicated each has some predictive value.
- Domain knowledge supports age, BMI, smoking, and family size as cost drivers.
- Region and sex, while weak, were included to avoid omitted variable bias.

3.2 Data Preprocessing

Categorical variables were converted using **one-hot encoding** (pandas get_dummies with drop_first=True to avoid dummy variable trap):

- Sex → sex_male (1 if male, else 0)
- Smoker → smoker_yes (1 if smoker, else 0)
- Region → region_northwest, region_southeast, region_southwest (northeast as baseline)

The final feature set includes 8 predictors: age, bmi, children, sex_male, smoker_yes and three region dummies.

3.3 Train-Test Split

- **Training set**: 80% of data (1,070 records)
- **Test set**: 20% of data (268 records)
- Random state = 42 ensures reproducibility.

3.4 Model Training

A LinearRegression model from scikit-learn was instantiated and fitted on the training data:
python
```
from sklearn.linear_model import LinearRegression
model = LinearRegression()
model.fit(X_train, y_train)
```
No feature scaling was applied because linear regression coefficients are scale-dependent but interpretable in original units.

3.5 Baseline Comparison (Added for Rigour)

A baseline model that always predicts the mean training charge (y_train.mean ()) yields an R^2 of **0** on the test set. Our model's R^2 of 0.78 represents a substantial improvement.

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4. Model Evaluation (20 Marks)

4.1 Evaluation Metrics

MAE	4181.19	On average, predictions deviate from actual charges by ~$4,181. Relative to mean charges ($13,270), this is a **31.5%** error.
MSE	33,596,915.85	Squared error is large due to outliers (high-cost smokers). This metric heavily penalises large errors.
R^2	0.7836	The model explains **78.4%** of the variance in insurance charges. Approximately 22% remains unexplained (due to missing predictors like medical history, or non-linear effects).

4.2 Comparison with Baseline

- **Baseline R^2 (predicting mean)** = 0.00
- **Our model R^2** = 0.78 → model is far better than a naïve guess.

4.3 Residual Analysis (Post-Evaluation)

A residual plot (predicted vs. actual) was examined (code not shown but recommended). Key observations:

- Residuals are roughly centred around zero, but show **heteroscedasticity** – variance increases with higher charges, confirming the impact of outliers.
- Normality of residuals is questionable due to skewness; a log transformation of charges could improve this.

4.4 Model Performance Summary

The model demonstrates **strong predictive power** ($R^2 > 0.75$) but has limitations:

- **MAE of $4,181** may be acceptable for risk segmentation but not for precise individual prediction.
- Outliers inflate MSE – consider robust regression or transformation.

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5. Interpretation of Results (20 Marks)

5.1 Coefficients

smoker_yes	+23,651	Smokers pay ~$23,650 more annually than non-smokers.
bmi	+337	Each 1-unit increase in BMI raises charges by $337.
age	+257	Each additional year adds $257 to annual charges.
children	+425	Each child adds $425 (small effect).

sex_male	**−19**	Males pay ~$19 less than females (negligible).
region_nortwest	**−371**	Northwest residents pay $371 less than Northeast (baseline).
region_southeast	**−658**	Southeast residents pay $658 less than Northeast.
region_southwest	**−810**	Southwest residents pay $810 less than Northeast.

5.2 Business Insights

- **Smoking is the dominant cost driver** – a premium loading of ~$23,650 for smokers is justified.
- **BMI and age** are secondary but significant – wellness programmes targeting weight reduction and preventive care for older adults could reduce costs.
- **Geographic variation** exists but is modest – regional pricing adjustments may be considered.
- **Sex and number of children** have minimal impact – they can be deprioritised in pricing models.

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6. Model Improvements (Optional but Recommended)

Although the model performs well, several enhancements could improve accuracy and robustness:

1. **Log transformation of charges –** to reduce skewness and stabilise variance, potentially increasing R^2.
2. **Interaction terms** – e.g., smoker * age, smoker * bmi – to capture synergistic effects.
3. **Outlier handling** – using Huber regression or trimming extreme values.
4. **Feature engineering** – create BMI categories (underweight, normal, overweight, obese) for non-linear effects.
5. **Regularisation** – Ridge or Lasso regression to prevent overfitting and perform feature selection.
6. **Advanced models** – Random Forest or Gradient Boosting to capture non-linear relationships.

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7. Conclusion

This project successfully developed a linear regression model to predict medical insurance charges using demographic and lifestyle data. The model explains 78% of the variance in charges, with smoking status identified as the most influential factor (coefficient ~$23,650). Age and BMI also contribute positively, while region and sex have minor effects.

Key takeaways for a medical aid scheme:

- **Risk-based pricing** should heavily penalise smokers.
- **Wellness incentives** targeting smoking cessation offer the highest potential return on investment.
- **Limitations** – the model misses ~22% of variance; adding medical history or using non-linear methods could improve predictions.

The model is ready for deployment as a baseline risk assessment tool, with clear recommendations for iterative improvement.

References

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